# Investigate a Suspicious File Hash: Pyramid of Pain

## Background

Security analysts investigate suspicious files to determine whether they are malicious and identify indicators that can help detect related activity. VirusTotal provides threat-intelligence information by collecting detection results and analysis from multiple security vendors and researchers.

The Pyramid of Pain explains different types of indicators of compromise (IoCs) and how difficult they can be for an attacker to change when defenders detect or block them. File hashes and IP addresses are examples of IoCs, while attacker tactics, techniques, and procedures (TTPs) can provide information about how an attack is carried out.

## Activity Overview

In this activity, I investigated a suspicious file associated with a phishing email at a financial services company. The employee opened a password-protected spreadsheet attachment, after which malicious executable files were created on the employee's computer.

I used the provided SHA-256 hash as the starting point for a VirusTotal investigation. The objective was to assess the file's reputation and identify additional indicators of compromise from the report's Detection, Details, Relations, and Behavior sections.

## Objective

- Assess whether the suspicious file is likely malicious.
- Review detection results from multiple security vendors.
- Identify additional IoCs associated with the file.
- Categorize the findings using the Pyramid of Pain.
- Record evidence to support further investigation and response.

## Tools Used

- **VirusTotal:** Used to review file reputation, vendor detections, related network indicators, and observed behavior.
- **Incident Handler's Journal:** Used to document the incident timeline and investigation notes.
- **Pyramid of Pain template:** Used to organize the identified IoCs by category.

## Incident Scenario

A financial services company received an alert after an employee opened a password-protected spreadsheet attached to an email. After the file was opened, multiple unauthorized executable files appeared on the employee's computer.

### Timeline

| Time | Event |
|---|---|
| 1:11 p.m. | Employee received an email with an attachment. |
| 1:13 p.m. | Employee downloaded and opened the file. |
| 1:15 p.m. | Multiple unauthorized executable files were created. |
| 1:20 p.m. | An intrusion detection system generated an alert for the SOC. |

# Initial Artifact

**Artifact type:** SHA-256 file hash

**Provided hash:**

```
54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93bab527f6b
```

This hash was used to search for the file in VirusTotal. A cryptographic hash acts as a file fingerprint, allowing analysts to look up reports associated with a particular file.

# Investigation Method

I reviewed the following sections of the VirusTotal report:

- **Detection:** To assess security vendor verdicts and the overall detection ratio.
- **Details:** To identify additional file hashes and static file information.
- **Relations:** To find related domains, IP addresses, and other connected artifacts.
- **Behavior:** To review sandbox observations, network activity, and techniques associated with the file.

A single VirusTotal result is not sufficient to establish every detail of an incident. Vendor detections, community feedback, and sandbox observations should be evaluated together.

# Assessment of the File

**Assessment:** The course exemplar identifies the file as malicious.

The exemplar states that more than 50 security vendors flagged the file and that multiple vendors associated it with Flagpro malware. This supports a malicious assessment in the context of the course activity.

**Verification note:** Before submitting my own worksheet, I should record the vendor detection ratio and community score shown in my VirusTotal report. The values can change over time, so I should not present unverified or outdated values as a current observation.

# Pyramid of Pain: Investigation Findings

The following examples are taken from the course exemplar. Confirm each item in the corresponding VirusTotal section before treating it as a verified finding from your investigation.

## 1. Hash Values

**Example IoC:** MD5 hash

```
287d612e29b71c90aa54947313810a25
```

**Source:** Details tab.

**Purpose:** Provides an additional file identifier that can be used to search for the same known file.

## 2. IP Addresses

**Example IoC:**

```
207.148.109.242
```

**Source:** Relations tab; the exemplar also associates this IP address with the domain in the DNS Resolutions section of a sandbox report.

**Purpose:** Can help identify or investigate network connections associated with the suspicious file. The IP address should be checked for context before being blocked because infrastructure may be shared or change ownership.

## 3. Domain Names

**Example IoC:**

```
org.misecure.com
```

**Source:** Relations tab.

**Purpose:** A domain reported as malicious in the course exemplar. It can be used for further investigation and, where appropriate, detection or blocking.

## Additional Pyramid of Pain Categories

The course exemplar also gives these examples:

**Network/host artifacts:** HTTP requests to `org.misecure.com`, as observed in sandbox reports.

**Tools:** Input capture, listed under the Collection tactic in the exemplar's sandbox findings. This describes a behavior or technique, not necessarily a named software tool.

**TTPs:** Command and Control, listed in the exemplar's sandbox findings. This describes an attacker tactic associated with communication between compromised systems and attacker-controlled infrastructure.

# Conclusion

This activity demonstrated how VirusTotal can help analysts assess a suspicious file and discover related indicators of compromise. Starting with a file hash, an analyst can investigate related domains, IP addresses, network activity, and attacker behaviors.

The Pyramid of Pain helps organize these findings and consider how useful different indicators may be for detection and defense. Findings should be verified against available evidence and used with other security tools when investigating a real incident.

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# Evidence and Limitations

The initial hash and incident timeline were supplied by the course scenario. The example IoCs in this report are drawn from the course exemplar and must be verified against the VirusTotal report used for the activity.

This report documents the investigation method and example findings; it does not claim that an independent malware analysis or sandbox execution was performed.